

School of Artificial, Electrical, Electronics and Computer Engineering
Department of Computer Engineering
CPE106L-4 | Software Design Laboratory

Project Proposal Paper

"Mapúa Findr: An Integrated Lost and Found Tracking Application "

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I. Background or Problem Statement

There are many students, faculty members, and staff in Mapúa University that might lose or recover personal property in the university grounds. The current way of reporting, logging, searching, and claiming lost and found items might be done manually through log books or by communicating with the person from campus. This method might cause difficulty in identifying and solving the problem, Mapúa Findr is introduced as an Integrated Lost and Found Tracking Application that would serve as a centralized platform for reporting, searching, validating, and claiming lost items. The system will have features for viewing and managing the logs of lost and found items of the authorized users while having facilities for tracking the location of the item, transfer of custody, requesting of claims, and status of the item.

II. Objectives

A centralised and authenticated Lost and Found Management System for Mapúa University, which is designed to facilitate and streamline the process of reporting, tracing, and reclaiming lost items within the university premises and ensure secure custody of objects throughout their life cycle.

- **Centralized Reporting & Database Management:**
    - To develop a single system that allows students, teachers, and administrators to report lost or surrendered items.
- **Search & Discovery:**
    - To develop a search and filter mechanism that will enable users to search for the found items using item name, category, campus location, and date found.
- **Authentication & Ownership Claims:**
    - To create a safe interface in which users will be able to make their claim applications and verify themselves using their Mapúa student credentials.
- **Custody Management & Accountability:**
    - To develop a fully functional custody tracking system that captures the physical location of storage, tracks all changes to custody of the items, and ensures a digital signature of acknowledgement each time custody is handed over.
- **Real-Time Automated Notifications:**

- To create an automated notification system that will provide feedback on the reported missing items as well as the processed claim requests through email or app notifications.

- **Administrative Audit Trails & Monitoring:**
  - To develop a unique dashboard for monitoring turnover of lost-and-found items, auditing digital trails, managing unclaimed items, and creating recovery reports for campus administrators.
- **Operational Efficiency:**
  - To reduce the total time, manual paperwork, and foot traffic on campus security desks for logging, searching for, verifying, and retrieving lost items.

III. Target Users

- **Students**
  - They can easily report their lost possessions, see live listings of found possessions on campus, and claim back their possessions through their school IDs.
- **Professors**
  - They can quickly surrender items left behind in lecture halls or laboratories to official desks, minimizing classroom disruptions and announcements during instructional time.
- **Admins**
  - They gain centralized visibility over campus lost-and-found operations, ensuring institutional accountability, policy enforcement, and compliance with data privacy standards.
- **Staff**
  - The System Custodians. They can replace manual paper logbooks with an electronic inventory system, monitor physical storage locations, and record item handovers.

IV. Scope and Limitations

The proposed Mapúa Findr: An Integrated Lost and Found Tracking Application will focus on developing a centralized web-based system for managing lost and found items within designated areas of Mapúa University. The system will provide authenticated access for students, professors, staff, and administrators according to their respective roles. It will allow users to report lost items, while authorized staff can record and manage items that have been found or surrendered. Each item record will contain relevant information such as its description, category, date, location, and current status. Users will also be able to search and filter found items and submit a claim for an item they believe belongs to them. Authorized staff will review and verify these claims, update item statuses, manage storage locations, record custody transfers, and document the release of claimed items. The system will also provide notifications and administrative records to help monitor the progress of lost and found items and maintain accountability throughout the process.

The proposed system will only cover lost and found items reported within designated areas of Mapúa University and will not be intended for use as a public lost and found platform. The tracking feature will be limited to recording the reported location, storage location, current status, and custody history of an item. It will not provide real time physical tracking through GPS, RFID, Bluetooth, or other tracking devices. The system will also not automatically determine whether a claimant is the rightful owner of an item. Ownership verification and the final decision to release an item will remain the responsibility of authorized university personnel. The accuracy of the records will depend on the information provided by users and staff. In addition, the initial version of the system will focus on the main lost and found process and will not replace existing university policies or necessarily integrate with all existing university information systems.

V. Tools and technologies

To fulfill the operational requirements and objectives of the Mapúa Findr application, the system will be developed and maintained utilizing the following technology stack:

- **Tkinter**: Builds the graphical user interface (GUI) for the desktop application, enabling intuitive navigation for students, staff, and campus administrators.

- **Python**: Handles backend system logic, database operations, custody tracking routines, automated notifications, and seamless communication across application modules.

- **MySQL**: Provides relational database storage for lost and found item records, physical storage locations, claim requests, and administrative audit trails.

- **Microsoft Entra ID**: Manages identity authorization and Single Sign-On (SSO) authentication, ensuring secure user login using Mapúa institutional credentials.

- **Visual Studio Code**: Serves as the primary Integrated Development Environment (IDE) for writing, debugging, and structuring the Python backend and Tkinter frontend codebase.

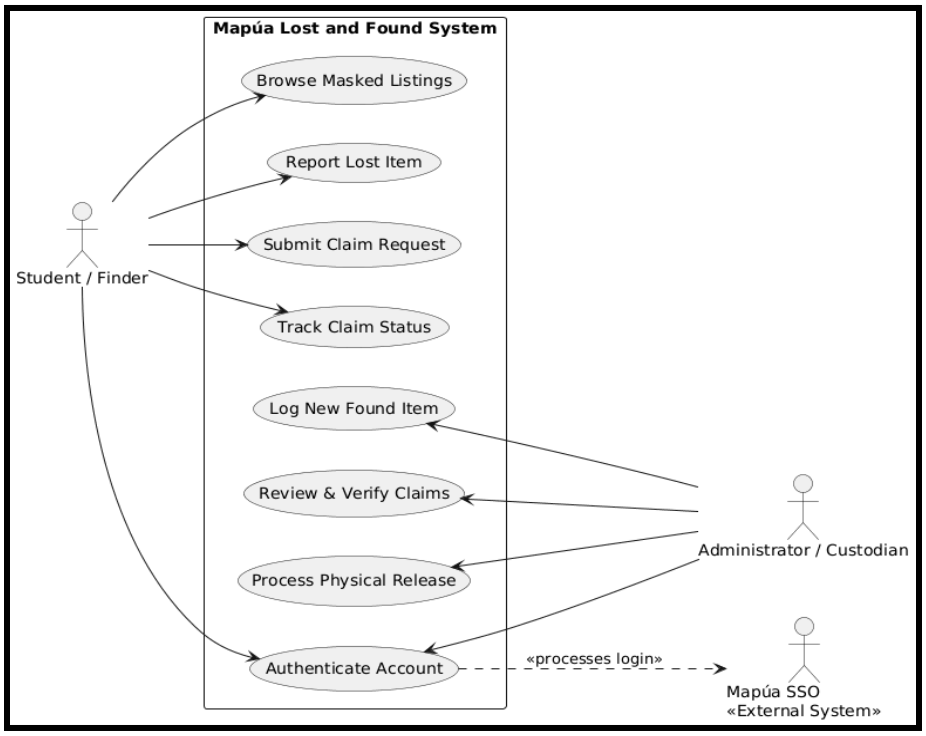
- **Git / GitHub**: Facilitates local and remote version control, collaborative code management, branch tracking, and repository storage.

VI. Initial Software Design Artifacts

A. Basic System Flow

- **Item Intake Flow:** Start → Finder Surrenders Item → Admin Logs Item & Hidden Details → System Generates QR Tag & Masked Public Listing → End
- **Item Claim Flow:** Start → Login (Mapúa SSO) → Browse Found Items → View Masked Details → Submit Claim Request (Blind Verification) → Admin Reviews Claim → Approval/Rejection Notification → Present QR Code for Physical Retrieval → Sign Digital Release → End.

B. Main Use Case



C. Initial Class Ideas

1. User Class (Base Class)

- **Description:** The parent class for all individuals accessing the platform, handling basic authentication via Mapúa's system.
- **Attributes:** user_id, name, mapua_email, user_role
- **Methods:** login(), logout()

2. Student Class (Inherits from User)

- **Description:** Represents the Mapúa students who will use the platform to search for their lost items and submit claim requests.
- **Attributes:** student_number, department
- **Methods:** viewListings(), submitClaimRequest(), trackClaimStatus(), reportLostItem()

3. Administrator Class (Inherits from User)

- **Description:** Represents the campus security or OSA personnel responsible for managing the physical inventory and verifying claims.
- **Attributes:** admin_level, assigned_desk_location
- **Methods:** logNewItem(), reviewClaim(), processItemRelease(), updateItemStatus()

4. FoundItem Class

- **Description:** Holds both the public and hidden descriptive data for a physical item surrendered to the campus authorities.
- **Attributes:** item_id, category, public_description, hidden_specifications, location_found, date_found, storage_bin, item_status (e.g., Active, Claimed, Archived)
- **Methods:** getMaskedDetails(), getFullDetails(), updateStatus()

5. ClaimRequest Class

- **Description:** Represents a transaction where a student attempts to verify ownership of a specific item.
- **Attributes:** claim_id, item_id, student_id, verification_answers (list), proof_image, claim_status (e.g., Pending, Action Required, Approved, Rejected), submission_date
- **Methods:** submitProof(), updateClaimStatus(), generateRetrievalQR()

6. Notification Class

- **Description:** Handles the automated alerts sent to users regarding their claim status or matching lost item reports.
- **Attributes:** notification_id, user_id, message_body, type (Email/Push), date_sent, is_read
- **Methods:** dispatchAlert(), markAsRead()

VII. Implementation Plan and Timeline

Week	Sprint Name	Key Tasks	Output
6	UI/ UX Design & Authentication Setup	● Design wireframes and high-fidelity layouts for the Student, Professor, Staff, and Admin dashboards.	● Functional Microsoft SSO login integration module

		• Build fronted view and interactive components for reporting lost items, searching inventory, and managing claims. • Implement Microsoft Entra ID Single Sign-On (SSO) integration to secure user login exclusively via official Mapúa credentials (@my.mapua.edu.ph).	• Complete frontend UI/UX wireframes and dashboard prototypes.
7	Database Schema & Role-Based Access	• Design relational database tables (Users, FoundItem, ClaimRequest, Notification, and AuditLogs) using MySQL. • Implement Role-Based Access Control (RBAC) to segment permissions across Students, Professors, Staff (Custodians), and Administrators. • Develop base CRUD operations for item logging, inventory categorization, and storage location mapping.	• Fully migrated and functional MySQL database schema • Role-secured backend routing structure
8	Core Logic & Custody Tracking	• Code the search and filter engine enabling users to look up items by name, category, campus location, and date found. • Program the claim verification workflow and the Custody Management tracking module (capturing storage bins, custody changes, and digital signatures for item handover).	• Working search filters and blind-verification claim submission module • Operational custody handover and digital log system

		• Integrate automated email or app notification mechanisms for missing items and claim statuses.	
9	System Integration & Testing	• Connect frontend components and user interfaces to the Python backend endpoints and MySQL database. • Perform comprehensive end-to-end testing across student reporting, custodian verification workflows, and admin audit dashboards. • Debug edge cases, handle access errors, and optimize system response times.	• Integrated, functional system prototype • Comprehensive system bug reports and test results
10	Finalization & Presentation	• Conduct User Acceptance Testing simulations with select campus security or student affairs personnel. • Prepare final project documentation, system user manuals, and source code repository (Github) handover. • Execute final system deployment and presentation defense.	• Deployment-Ready Web/Desktop Application • Final Defense Presentation and Documentation.